

# A cautionary tale on the efficiency of some adaptive Monte Carlo Schemes

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**Abstract:** There is a growing interest in the literature for adaptive Markov Chain Monte Carlo methods based on sequences of random transition kernels  $\{P_n\}$  where  $P_n$  is permitted to have an invariant distribution  $\pi_n$  not necessarily the target distribution  $\pi$ . These algorithms are designed such that as  $n \rightarrow \infty$ ,  $P_n$  converges to  $P$  a transition kernel with invariant distribution  $\pi$ . The interest of the approach is that typically,  $P$  is a very efficient kernel that would be difficult to implement otherwise. It is thought that as  $n \rightarrow \infty$  the algorithm will inherit the nice properties of the limiting kernel  $P$ . The *Equi-Energy sampler* of [13] is one popular example. Another example is the *Importance Resampling* MCMC ([5], [2]). The purpose of this paper is to show that, this type of samplers do not behave in general in accordance with the limiting kernel  $P$  and can sometimes lead to inefficient samplers. We show this through a simulation example and more rigorously by deriving a central limit theorem for the *Importance Resampling* MCMC sampler.

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## 1. Introduction

Adaptive Markov Chain Monte Carlo (MCMC) is a growing approach to Monte Carlo simulation where the transition kernel of the algorithm is allowed to change over time as an attempt to improve efficiency. It grows out of the seminal works of [9], [4], [11]. The general idea can be presented as follows. Let  $\pi$  be the distribution of interest. Let  $\{P_\theta\}$  be a family of transition kernels each having  $\pi$  as invariant distribution. The problem is to sample the most efficiently from  $\pi$  using the family  $\{P_\theta\}$ . This can be solved adaptively: given all the information available at time  $n$ , we pick a new “best” kernel  $P_{\theta_n}$  and generate  $X_{n+1}$  from  $P_{\theta_n}(X_n, \cdot)$ . Some general sufficient conditions for the consistency of such algorithms can be found in [6], [3], [15]. These conditions have been shown to hold for many algorithms used in practice. Consistency here means that the distribution of  $X_n$  indeed converges to  $\pi$ . It is also shown in [1] under some regularity conditions that, if a “best” limiting kernel  $P_{\theta^*}$  is found, the adaptive chain  $\{X_n\}$  behaves essentially like a standard Markov chain with transition kernel  $P_{\theta^*}$ . In all the above mentioned papers, it is always assumed that each  $P_\theta$  has invariant distribution  $\pi$ .

More recently, interest has emerged in building Monte Carlo algorithms where the transition kernel  $P_n$  used at time  $n$  has invariant distribution  $\pi_n$  not necessarily equal to  $\pi$ . These algorithms are designed such that as  $n \rightarrow \infty$ ,  $P_n$  converges to a transition kernel  $P$  which is invariant with respect to  $\pi$ . This limiting kernel  $P$  is typically a very efficient kernel that would be extremely difficult to implement otherwise. Clearly, the interest of the method is that as  $n \rightarrow \infty$ ,  $P_n$  approaches  $P$  and one expects the algorithm to inherit the efficiency of  $P$  and perform like a classical MCMC algorithm with transition kernel  $P$ . The Equi-Energy sampler of [13] is an example. [13] gives numerous simulation problems where the Equi-Energy sampler outperforms alternative

methods. Another example based on importance resampling appeared independently in [5] and [2]. We call it here *importance resampling MCMC* (IR-MCMC). Some general consistency results on the Equi-Energy sampler can be found in [2]. This paper addresses the question of efficiency of this new Monte Carlo approach. That is, whether such algorithms perform in accordance with their limiting kernel  $P$  as one might first think.

We show that contrary to the case of classical adaptive MCMC briefly outlined above, and quite surprisingly, the behavior of this new type of adaptive MCMC samplers can be quite different from the behavior of their limiting kernel  $P$ . This is done by analyzing the IR-MCMC algorithm. We give an example where IR-MCMC and Equi-Energy sampler are outperformed by a plain Random Walk Metropolis although their limiting kernels are shown to be largely superior to this latter algorithm. More rigourously, we derive a central limit theorem for additive functionals of the IR-MCMC process and compare it with the asymptotics of its limiting kernel. We find that the discrepancy between  $\pi_n$  and  $\pi$  actually introduces an additional bias term (typically of the order  $1/\sqrt{n}$ ) that contributes to (and often inflates) the asymptotic variance of IR-MCMC. Although we do not study the asymptotics of the Equi-Energy sampler, our simulation results show that the Equi-Energy sampler also suffers from the same problem. Our main point is that the performance of this new type of Monte Carlo sampler cannot be easily predicted. And in order to reap its full potential, a more subtle approach to learning the limiting kernel  $P$  is needed.

Another contribution of the paper is that it gives a law of large numbers and the central limit theorem for additive functionals of the IR-MCMC process. The proof is based on the martingale approximation technique (see e.g. [12] and [14] for more background details including historical notes). See [3] for an earlier application to adaptive MCMC processes. A distinctive feature here is the fact that the IR-MCMC process is a set of parallel and self-interacting processes with

an hierarchy between the processes. We show that in this case the martingale approximation propagates: if the  $(l - 1)$ -th chain in the hierarchy has a martingale approximation then the  $l$ -th chain also has a martingale approximation. The approach can be extended to analyze similar self-interacting processes.

The paper is organized as follows. In Section 2 we present the IR-MCMC algorithm and discuss its efficiency. The case of the equi-energy sampler is discussed in 3. The limit theorems are developed in 4.

## 2. The Importance-Resampling MCMC algorithm and its efficiency

This section discusses the efficiency of Importance-Resampling MCMC (IR-MCMC). The algorithm is a natural and very appealing Monte Carlo scheme where empirical measure approximations of the target distribution are used to try to speed up convergence.

### 2.1. The Importance-Resampling MCMC sampler

Let  $\pi(dx) \propto h(x)\lambda(dx)$  be a distribution of interest on a states space  $(\mathcal{X}, \mathcal{B}, \lambda)$ . Typically  $\mathcal{X} \subseteq \mathbb{R}^d$ ,  $d$  large;  $\lambda$  is the Lebesgue measure and  $h$  is some complicated nonnegative measurable function. Let  $\pi^{(0)}(dx) \propto h^{(0)}(x)\lambda(dx)$  be some distribution on  $(\mathcal{X}, \mathcal{B}, \lambda)$  that is easy to sample from. Here “easy to sample from” means that either a fast exact algorithm or a fast mixing MCMC algorithm is available to sample from  $\pi^{(0)}$ . It is often helpful, but not necessary, to have  $\pi^{(0)}$  close to  $\pi$ . In this context, a powerful and well-known Monte Carlo strategy (*tempering*) consists in interpolating  $\pi^{(0)}$  and  $\pi$  through a sequence of distributions  $\{\pi^{(l)}, l = 0, \dots, K\}$ , where  $\pi^{(l)}(dx) \propto h^{(l)}(x)\lambda(dx)$  and  $\pi^{(K)} = \pi$ . Samples generated from the “easy” *tempered* distributions  $\pi^{(l)}$  can help sample more efficiently from  $\pi$ .

To describe the IR-MCMC algorithm, we assume that a transition kernel  $P^{(l)}$  is available that has invariant distribution  $\pi^{(l)}$  and denote  $\omega_l(x) = h^{(l)}(x)/h^{(l-1)}(x)$ . Finally, let  $T^{(l)}$  be a transition kernel on  $(\mathcal{X}, \mathcal{B})$  that also has invariant distribution  $\pi^{(l)}$ . The algorithm builds a parallel self-interacting chain  $\{(X_n^{(0)}, \dots, X_n^{(K)}), n \geq 0\}$  on  $\mathcal{X}^{K+1}$  as follows. We start with some initial values  $(X_0^{(0)}, \dots, X_0^{(K)}) \in \mathcal{X}^{K+1}$ . Then at time  $n$ , given the  $\sigma$ -algebra generated by  $\{(X_k^{(0)}, \dots, X_k^{(K)}), k \leq n\}$ , we do the following. We move  $\{X_n^{(0)}\}$  as a standard MCMC algorithm:  $X_{n+1}^{(0)} \sim P^{(0)}(X_n^{(0)}, \cdot)$ . For  $l = 1, \dots, K$ , we generate a random variable  $Y^{(l)}$  by resampling from  $\{X_0^{(l-1)}, \dots, X_n^{(l-1)}\}$  according to weights  $\{\omega_0^{(l)}, \dots, \omega_n^{(l)}\}$ , where  $\omega_k^{(l)} = \omega_l(X_k^{(l-1)})$ . Then with probability  $\theta_l$ , we sample  $X_{n+1}^{(l)}$  from  $P^{(l)}(X_n^{(l)}, \cdot)$  and with probability  $1 - \theta_l$ , we sample  $X_{n+1}^{(l)}$  from  $T^{(l)}(Y^{(l)}, \cdot)$ .

**Algorithm 2.1.** At time  $n$ , given  $\{(X_k^{(0)}, \dots, X_k^{(K)}), k \leq n\}$ :

1. Sample  $X_{n+1}^{(0)} \sim P^{(0)}(X_n^{(0)}, \cdot)$ .
2. For  $l = 1, \dots, K$  and  $\theta_l \in (0, 1)$ , Sample  $X_{n+1}^{(l)}$  from  $S_n^{(l)}(X_n^{(l)}, \cdot)$ , where

$$S_n^{(l)}(x, A) = \theta_l P^{(l)}(x, A) + (1 - \theta_l) \frac{\sum_{k=1}^n \omega_k^{(l)} T^{(l)}(X_k^{(l-1)}, A)}{\sum_{k=1}^n \omega_k^{(l)}},$$

where  $\omega_k^{(l)} = \omega_l(X_k^{(l-1)})$ . In other words, with probability  $\theta_l$  we sample  $X_{n+1}^{(l)}$  from  $P^{(l)}(X_n^{(l)}, \cdot)$

and with probability  $1 - \theta_l$ , we obtain  $Y^{(l)}$  by resampling from  $\{X_0^{(l-1)}, \dots, X_n^{(l-1)}\}$  with

weights  $\{\omega_k^{(l)}, k \leq n\}$  and then propose  $X_{n+1}^{(l)} \sim T^{(l)}(Y^{(l)}, \cdot)$ .

### 2.1.1. Heuristics

Clearly  $\{X_n^{(0)}\}$  is a Markov chain with stationary distribution  $\pi^{(0)}$ . Thus for  $n$  large enough,  $\{X_k^{(0)}, k \leq n\}$  can be seen as a sample from  $\pi^{(0)}$  and  $\sum_{k=1}^n \omega_k^{(1)} T^{(1)}(X_k^{(0)}, \cdot) / \sum_{k=1}^n \omega_k^{(1)}$  an empirical measure estimate of  $\pi^{(1)}$ . This means that for  $n$  large,  $S_n^{(1)}(x, A) = \theta_1 P^{(1)}(x, A) +$

$(1 - \theta_1) \sum_{k=1}^n \omega_k^{(1)} T^{(1)}(X_k^{(0)}, A) / \sum_{k=1}^n \omega_k^{(0)}$  should behave like  $S^{(1)}(x, A) = \theta_1 P^{(1)}(x, A) + (1 - \theta_1) \pi^{(1)}(A)$ . This makes the algorithm particularly appealing: for  $n$  large, each importance resampling should give an almost perfect sampling from  $\pi^{(1)}$ . Now, once  $X_n^{(1)}$  converges to  $\pi^{(1)}$ , the same heuristical argument just developed also applies and justified that  $\{X_n^{(2)}\}$  should be sampling efficiently from  $\pi^{(2)}$ , and so on.

Typically,  $P^{(l)}$  will be a Metropolis-Hastings kernel or a Gibbs kernel with invariant distribution  $\pi^{(l)}$ .  $T^{(l)}$  can be any transition kernel (not necessarily ergodic) that is invariant with respect to  $\pi^{(l)}$ . The typical choice is  $T^{(l)} = I$ , the identity kernel. The choice of  $\theta_l$  overall should depend on the quality of the interpolation. When  $\pi^{(0)}$  and  $\pi$  are well interpolated, the resampling step should work well and one can simply choose  $\theta_l = 0.5$ .

As with any *importance sampling* method, we can check the quality of the interpolation  $\{\pi^{(l)}\}$  by computing for each  $l = 1, \dots, K$ , one plus the variance of the weight function  $\omega_l$ :

$$\text{eff}^{(l)} = 1 + \mathbb{V}_{\pi^{(l-1)}}(\tilde{\omega}_l(X)), \quad (1)$$

where  $\tilde{\omega}_l(x) = \omega_l(x) / \mathbb{E}_{\pi^{(l-1)}}(\omega_l(X))$ . Large values of  $\text{eff}^{(l)}$  indicates a high variability in the importance weights which typically means a high discrepancy between  $\pi^{(l-1)}$  and  $\pi^{(l)}$ . A consistent estimator of  $\text{eff}^{(l)}$  is given by:

$$\text{eff}_n^{(l)} = \frac{n \sum_{k=1}^n \omega_l^2(X_k^{(l-1)})}{\left\{ \sum_{k=1}^n \omega_l(X_k^{(l-1)}) \right\}^2}. \quad (2)$$

In practice, one can use  $\text{eff}^{(l)}$  to construct interpolating distributions of the form  $\pi^{(l)} = \pi^{1/t_l}$ : based on a small pilot simulation, we choose  $t_1$  as the largest “temperature”  $t$  for which  $\text{eff}^{(1)}$  remains smaller than some threshold (say 2). Given  $t_1$ , we choose  $t_2$  as the largest  $t$  for which  $\text{eff}^{(2)}$  remains smaller than the threshold. And so on. We stop once  $\pi^{1/t_K}$  looks sufficiently flat.

## 2.2. A toy example

Consider the following simple problem. Suppose that we want to sample from the bivariate normal distribution  $\mathcal{N}(0, \Sigma)$ , with covariance matrix

$$\Sigma = \begin{bmatrix} 0.96 & 2.44 \\ 2.44 & 7.04 \end{bmatrix}.$$

We use this simple sampling problem to compare a Random Walk Metropolis (RWM) algorithm, IR-MCMC and the MCMC algorithm with transition kernel given by the limit kernel of IR-MCMC (that we call here the limit IR-MCMC sampler).

For the RWM sampler, the proposal kernel is  $\mathcal{N}(0, I_2)$ , where  $I_2$  is the 2-dimensional identity matrix. For IR-MCMC, we use four chains with  $\pi^{(0)} = \pi^{1/10}$ ,  $\pi^{(1)} = \pi^{1/5}$ ,  $\pi^{(2)} = \pi^{1/2}$  and  $\pi^{(3)} = \pi$ . We take  $\theta_l = \theta = 0.5$  and  $P^{(l)}$  is taken to be a RWM algorithm with target  $\pi^{(l)}$  and proposal  $\mathcal{N}(0, I_2)$ . The limit IR-MCMC sampler has transition kernel  $\theta P^{(3)} + (1 - \theta)\pi$ . We simulate each of the three samplers for  $N = 10,000$  iterations. In order to estimate the accuracy of the methods, we repeat the simulations 100 times. The mean square errors (MSE) in estimating the two first moments of the two components of the distribution are reported in Table 1. From these results we see (as expected) that the limit IR-MCMC sampler is 15 to 30 times more efficient than the RWM sampler. Whereas IR-MCMC itself is hardly more efficient than the RWM sampler. There is a considerable loss of efficiency going from the limit kernel to the actual IR-MCMC. Now, if we take the computation times into account, it is clear that IR-MCMC is in any way more efficient than the RWM.

It is important to note that this poor performance of IR-MCMC is not due to a poor choice of the interpolating distributions  $\pi^{(t)}$ . With the choice given above, the estimated efficiencies  $\text{eff}_n^{(l)}$

are all smaller than 2.

|               |        | $\mathbb{E}(X_1)$ | $\mathbb{E}(X_2)$ | $\mathbb{E}(X_1^2)$ | $\mathbb{E}(X_2^2)$ |
|---------------|--------|-------------------|-------------------|---------------------|---------------------|
| RWM           | MSE    | 0.0099            | 0.0803            | 0.0091              | 0.5525              |
|               | Ratios | 1.0               | 1.0               | 1.0                 | 1.0                 |
| IR-MCMC       | MSE    | 0.0098            | 0.0774            | 0.0047              | 0.2962              |
|               | Ratios | 1.00              | 1.04              | 1.95                | 1.87                |
| limit IR-MCMC | MSE    | 0.0002            | 0.0017            | 0.0006              | 0.0296              |
|               | Ratios | 48.43             | 46.20             | 14.18               | 18.66               |

TABLE 1

Mean square error and ratios (with respect to the RWM sampler) for IR-MCMC and limit IR-MCMC. Based on 100 replications of 10,000 iterations of each sampler.

### 2.3. Understanding the asymptotics of IR-MCMC

As discussed heuristically above, as  $n \rightarrow \infty$ , the sequence of transition kernels in the  $l$ -th chain of IR-MCMC approaches  $\theta_l P^{(l)}(x, A) + (1 - \theta_l)\pi^{(l)}(A)$  which is clearly a much more efficient kernel than  $P^{(l)}$ . Nevertheless, the algorithm can sometimes perform relatively poorly. To understand these results, we study the rate of convergence of empirical averages of the form  $\sum_{k=1}^n f(X_k^{(l)})/n$ . Before stating the results, we introduce some standard notations. Denote  $\Pr$  the underlying probability measure and  $\mathbb{E}$  its expectation operator. We define the multiplication of two transition kernels:  $P_1 P_2(x, A) := \int P_1(x, dy) P_2(y, A)$ . A transition kernel  $P_1$  induces a linear operator (also denoted  $P_1$ ) on the space of  $\mathbb{R}$ -valued functions on  $(\mathcal{X}, \mathcal{B})$  by  $P_1 f(x) := \int P_1(x, dy) f(y)$ . If  $\mu$  is a signed measure on  $(\mathcal{X}, \mathcal{B})$ , we denote  $\mu(f) := \int \mu(dx) f(x)$  and we will also write  $\mu$  to denote the linear functional on the space of  $\mathbb{R}$ -valued functions on  $(\mathcal{X}, \mathcal{B})$  thus induced. Finally, we define  $\mu P_1(A) := \int \mu(dx) P_1(x, A)$ . Let  $V : \mathcal{X} \rightarrow [1, \infty)$  be given. For  $f : \mathcal{X} \rightarrow \mathbb{R}$ , we define its  $V$ -norm  $|f|_V := \sup_{x \in \mathcal{X}} \frac{|f(x)|}{V(x)}$  and we introduce the space  $L_V := \{f : \mathcal{X} \rightarrow \mathbb{R} : |f|_V < \infty\}$ .  $(L_V, |\cdot|_V)$  is a Banach space. For a signed measure  $\mu$  on  $(\mathcal{X}, \mathcal{B})$  we define by  $\|\mu\|_V := \sup_{f \in L_V, |f|_V \leq 1} |\mu(f)|$  the

norm of the linear functional  $\mu$  restricted on  $(L_V, |\cdot|_V)$ . Similarly, for a linear operator  $T$  from  $(L_V, |\cdot|_V)$  into itself, we define its operator norm by  $\|T\|_V := \sup_{f \in L_V, \|f\|_V \leq 1} \|Tf\|_V$ .

**Assumption (A):**

**(A1)** *There exist  $0 < c < C < \infty$  such that  $c \leq \omega_l(x) \leq C$ , for all  $x \in \mathcal{X}$  and  $l = 1, \dots, K$ .*

**(A2)**  *$P^{(0)}$  is  $\phi$ -irreducible, aperiodic and there exists a measurable function  $V : \mathcal{X} \rightarrow [1, \infty)$ , such that*

$$\left\| \left[ P^{(0)} \right]^n - \pi^{(0)} \right\|_V = O(\rho_0^n). \quad (3)$$

**(A3)** *For any  $\beta \in [0, 1]$ ,*

$$\left| T^{(l)} V^\beta \right|_{V^\beta} + \sup_{i \geq 0} \left| [P^{(l)}]^i V^\beta \right|_{V^\beta} < \infty, \quad (4)$$

*for all  $l = 0, \dots, K$ , where  $V$  is as in (A2).*

**Remark 2.1.** 1. In (A1), we assume that  $\omega_l$  is bounded and bounded away from 0. Assuming  $\omega_l$  bounded is a natural and important assumption to assure the stability of the resampling weights. But the assumption that  $\omega_l$  is bounded away from 0 is of a technical nature and is introduced mainly to simplify the proofs.

2. In (A2), we require that  $P^{(0)}$  converges to  $\pi^{(0)}$  at a geometric rate. This is not a particular strong assumption since we control to some extent the choice of  $\pi^{(0)}$  and  $P^{(0)}$ .

3. It is also worth noting that we do not make any specific assumption about the convergence rate of the original transition kernel  $P^{(l)}$ ,  $l \geq 1$ . We only require that  $\left| [P^{(l)}]^n V^\beta \right|_{V^\beta}$  be bounded in  $n$ . In most cases we take  $T^{(l)}$  to be the identity kernel or  $P^{(l)}$  in which cases  $\left| T^{(l)} V^\beta \right|_{V^\beta} < \infty$  trivially hold.

**Theorem 2.1.** Assume that (A) hold and fix  $l \in \{0, \dots, K\}$ .

i. If  $f \in L_{V^\beta}$  for some  $\beta \in [0, 1/4)$  then

$$\frac{1}{n} \sum_{i=1}^n f(X_i^{(l)}) \longrightarrow \pi^{(l)}(f), \text{ Pr} - a.s. \text{ as } n \rightarrow \infty. \quad (5)$$

ii. Let  $f \in L_{V^\beta}$  for some  $\beta \in [0, 1/8)$  such that  $\pi^{(l)}(f) = 0$ . Define

$$\sigma_l^2(f) := \pi^{(l)}(f^2) + 2 \sum_{k=1}^{\infty} \theta_l^k \pi^{(l)} \left[ \left[ P^{(l)} \right]^k f \right], \quad (6)$$

and assume that  $\sigma_l^2(f) > 0$ . Then there exists a random sequence  $\{\pi_n^{(l)}(f)\}$ ,  $\pi_n^{(l)}(f) \rightarrow \pi^{(l)}(f)$

(almost surely) as  $n \rightarrow \infty$  such that:

$$\frac{1}{\sqrt{n}\sigma_l(f)} \sum_{k=1}^n \left[ f(X_k^{(l)}) - \pi_k^{(l)}(f) \right] \Rightarrow \mathcal{N}(0, 1) \quad \text{as } n \rightarrow \infty, \quad (7)$$

where  $\Rightarrow$  stands for the convergence in distribution.

*Proof.* See Section 4. □

Denote  $S^{(l)}(x, A) = \theta_l P^{(l)}(x, A) + (1 - \theta_l) \pi^{(l)}(A)$  the  $l$ -th limiting kernel of IR-MCMC. Let  $f$  as in the theorem. Fix  $l$  and assume that  $\pi^{(l)}(f) = 0$ . It is interesting to note that  $\sigma_l^2(f)$  is precisely the asymptotic variance for  $f$  in the central limit theorem for the Markov chain with transition kernel  $S^{(l)}$ . More precisely, if  $\{Y_n^{(l)}\}$  is a Markov chain with transition kernel  $S^{(l)}$ , it is easy to show that  $\{Y_n^{(l)}\}$  is geometrically ergodic and that  $\sum_{k=1}^n f(Y_k^{(l)})/n$  satisfies a central limit theorem:

$$\frac{1}{\sqrt{n}\sigma_l(f)} \sum_{k=1}^n f(Y_k^{(l)}) \Rightarrow \mathcal{N}(0, 1),$$

where  $\sigma_l^2(f)$  is given by (6).

Now, from Theorem 2.1 we deduce that for  $n$  large:

$$\frac{1}{n} \sum_{k=1}^n f(X_k^{(l)}) = \frac{1}{n} \sum_{k=1}^n \pi_k^{(l)}(f) + \varepsilon_n^{(l)}, \quad (8)$$

where

$$\varepsilon_n^{(l)} \approx \mathcal{N}\left(0, \frac{\sigma_l^2(f)}{n}\right). \quad (9)$$

Therefore the precision of the Monte Carlo estimate  $\sum_{k=1}^n f(X_k^{(l)})/n$  is

$$\mathbb{E}\left(\frac{1}{n} \sum_{k=1}^n f(X_k^{(l)})\right)^2 \approx \frac{\sigma_l^2(f)}{n} + \mathbb{E}\left(\frac{1}{n} \sum_{k=1}^n \pi_k^{(l)}(f)\right)^2 + 2\mathbb{E}\left(\frac{1}{n} \sum_{k=1}^n \pi_k^{(l)}(f) \varepsilon_n^{(l)}\right).$$

Now we see the problem with  $\sum_{k=1}^n f(X_k^{(l)})/n$ : the term  $\frac{1}{n} \sum_{k=1}^n \pi_k^{(l)}(f)$  in (8) is not negligible in general. As we show below in the proofs,

$$\pi_n^{(l)}(f) = \mu_n \left( (1 - \theta_l) \sum_{i=0}^{\infty} \theta_l^i [P^{(l)}]^i f \right),$$

where

$$\mu_n^{(l)}(f) = \frac{\sum_{k=1}^n \omega_k^{(l)} T^{(l)} f(X_k^{(l-1)})}{\sum_{k=1}^n \omega_k^{(l)}}.$$

As an empirical average of an ergodic process,  $\mu_n^{(l)}(f)$  is typically of the order  $1/\sqrt{n}$  which implies that  $\pi_n^{(l)}(f)$  and  $\frac{1}{n} \sum_{k=1}^n \pi_k^{(l)}(f)$  are also of the order  $1/\sqrt{n}$ . So the efficiency of  $\sum_{k=1}^n f(X_k^{(l)})/n$  in general will not be determined by  $\sigma_l^2(f)$  alone. It is possible to have  $\mathbb{E}\left(\frac{1}{n} \sum_{k=1}^n f(X_k^{(l)})\right)^2 < \sigma_l^2(f)/n$  due to the covariance term but more likely and as in the simulation example above, we will have  $\mathbb{E}\left(\frac{1}{n} \sum_{k=1}^n f(X_k^{(l)})\right)^2 \geq \sigma_l^2(f)/n$ .

### 3. Extension to the Equi-Energy sampler

Equi-Energy (EE) sampling suffers from the same problem. This algorithm shares the same structure with IR-MCMC but uses a different strategy. Define  $E(x) = -\log(\pi(x))$  the energy function. Let  $\{\mathcal{X}_i, i = 1, \dots, d\}$  be a partition of  $\mathcal{X}$  where  $\mathcal{X} = \{x \in \mathcal{X} : E_{i-1} < E(x) \leq E_i\}$  for some predefined values  $E_0 < E_1 \dots < E_d$ . Define the function  $I(x) = i$  if  $x \in \mathcal{X}_i$ ; so  $\mathcal{X}_{I(x)}$  represents the component of the partition to which  $x$  belongs. The EE sampler also uses the

idea of tempering. As with the IR-MCMC, let  $\{\pi^{(l)}, l = 0, \dots, K\}$  be a sequence of distribution such that  $\pi^{(l)}(dx) \propto h^{(l)}(x)\lambda(dx)$  and  $\pi^{(K)} = \pi$  and let  $P^{(l)}$  be a transition kernel with invariant distribution  $\pi^{(l)}$ . As before, write  $\omega_l(x) = h^{(l)}(x)/h^{(l-1)}(x)$  the importance functions. The Equi-Energy sampler is a multiple-chains sampler that works as follows:

**Algorithm 3.1.** At time  $n$ , given  $\{(X_k^{(0)}, \dots, X_k^{(K)}), k \leq n\}$ :

1. Sample  $X_{n+1}^{(0)} \sim P^{(0)}(X_n^{(0)}, \cdot)$ .
2. For  $l = 1, \dots, K$  and  $\theta_l \in (0, 1)$ , Sample  $X_{n+1}^{(l)}$  from  $K_n^{(l)}(X_n^{(l)}, \cdot)$ , where

$$K_n^{(l)}(x, A) = \theta_l P^{(l)}(x, A) + (1 - \theta_l) \left\{ \frac{\sum_{k=1}^n \min\left(1, \frac{\omega_l(X_k^{(l-1)})}{\omega_l(x)}\right) \mathbf{1}_{A \cap \mathcal{X}_{I(x)}}(X_k^{(l-1)})}{\sum_{k=1}^n \mathbf{1}_{A \cap \mathcal{X}_{I(x)}}(X_k^{(l-1)})} + (1 - a_n^{(l)}(x)) \mathbf{1}_A(x) \right\},$$

where  $a_n^{(l)}(x) = \sum_{k=1}^n \min\left(1, \frac{\omega_l(X_k^{(l-1)})}{\omega_l(x)}\right) \mathbf{1}_{A \cap \mathcal{X}_{I(x)}}(X_k^{(l-1)}) / \sum_{k=1}^n \mathbf{1}_{A \cap \mathcal{X}_{I(x)}}(X_k^{(l-1)})$ . When the set  $\{X_k^{(l-1)} : X_k^{(l-1)} \in \mathcal{X}_{I(X_n^{(l)})}, k \leq n\}$  is empty and a resampling is required, we stay put at  $X_n^{(l)}$ .

In other words, with probability  $\theta_l$  we sample  $X_{n+1}^{(l)}$  from  $P(X_n^{(l)}, \cdot)$  and with probability  $1 - \theta_l$ , we obtain  $Y^{(l)}$  by resampling uniformly from  $\{X_k^{(l-1)} : X_k^{(l-1)} \in \mathcal{X}_{I(X_n^{(l)})}, k \leq n\}$ . Then  $Y^{(l)}$  is accepted with probability  $\min\left(1, \frac{\omega_l(Y^{(l)})}{\omega_l(X_n^{(l)})}\right)$  in which case we set  $X_{n+1}^{(l)} = Y^{(l)}$ ; otherwise  $Y^{(l)}$  is rejected and we set  $X_{n+1}^{(l)} = X_n^{(l)}$ .

As  $n$  approach infinity,  $K_n^{(l)}$  approach the mixture of transition kernels given by:

$$K^{(l)}(x, A) = \theta_l P^{(l)}(x, A) + (1 - \theta_l) R^{(l)}(x, A), \quad (10)$$

where  $R^{(l)}$  is a Metropolis-Hastings algorithm with target distribution  $\pi^{(l)}$  and proposal kernel  $Q^{(l)}(x, dy) \propto h^{(l-1)}(y) \mathbf{1}_{\mathcal{X}_{I(x)}}(y) \lambda(dy)$ .

To help the reader see the analogy between IR-MCMC and EE sampling, assume for a moment that there is only one component in the partition. Then  $K^{(l)}(x, A) = \theta_l P^{(l)}(x, A) + (1 - \theta_l)R^{(l)}(x, A)$  becomes a mixture of  $P^{(l)}$  and an Independent-Metropolis sampler with proposal  $\pi^{(l-1)}$ , while the limit kernel of the IR-MCMC is  $S^{(l)}(x, A) = \theta_l P^{(l)}(x, A) + (1 - \theta_l)\pi^{(l)}$  a mixture of  $P^{(l)}$  and a perfect sampling from  $\pi^{(l)}$ . Intuitively,  $S^{(l)}$  seems better than  $K^{(l)}$  and this can be proved rigorously in some cases. For example if, as in the example treated above,  $\pi^{(l)}(dx) \propto e^{-E(x)/t_l}\lambda(dx)$  with  $t_l < t_{l-1}$  and  $E(x) \geq 0$  then  $S^{(l)}$  is always better than  $K^{(l)}$  in terms of asymptotic variances. This follows from Peskun's Theorem ([16]) and the fact that for  $x \notin A$ :

$$\begin{aligned} S^{(l)}(x, A) &= K^{(l)}(x, A) + (1 - \theta_l) \left[ \pi^{(l)}(A) - R^{(l)}(x, A) \right] \\ &= K^{(l)}(x, A) + (1 - \theta_l) \left[ \pi^{(l)}(A) - \int_A \min \left( \frac{1}{\omega_l(x)}, \frac{1}{\omega_l(y)} \right) \pi^{(l)}(dy) \right] \\ &\geq K^{(l)}(x, A). \end{aligned}$$

But of course, the idea of partitioning the state space and using the proposal  $Q^{(l)}(x, dy) \propto h^{(l-1)}(y) \mathbf{1}_{\mathcal{X}_{I(x)}}(y) \lambda(dy)$  is a powerful idea that gives a tremendous edge to the EE sampler for more complicated sampling problems where in general the distance between  $\pi^{(l)}$  and  $\pi^{(l-1)}$  makes the Importance-Resampling step in IR-MCMC often inefficient.

### 3.1. Efficiency of the Equi-Energy sampler

EE sampling suffers from the same problem as IR-MCMC. This see that, we reproduce the simulations of Example 2.2 but this time comparing the RWM sampler, the EE sampler and the MCMC sampler with transition kernel given by the limit kernel of the EE sampler (we call it the limit EE sampler). As above we use  $\pi^{(0)} = \pi^{1/10}$ ,  $\pi^{(1)} = \pi^{1/5}$ ,  $\pi^{(2)} = \pi^{1/2}$  and  $\pi^{(3)} = \pi$  and

|          |        | $\mathbb{E}(X_1)$ | $\mathbb{E}(X_2)$ | $\mathbb{E}(X_1^2)$ | $\mathbb{E}(X_2^2)$ |
|----------|--------|-------------------|-------------------|---------------------|---------------------|
| RWM      | MSE    | 0.0099            | 0.0803            | 0.0091              | 0.5525              |
|          | Ratios | 1.0               | 1.0               | 1.0                 | 1.0                 |
| EE       | MSE    | 0.0057            | 0.0435            | 0.0045              | 0.2810              |
|          | Ratios | 1.74              | 1.84              | 2.02                | 1.97                |
| limit EE | MSE    | 0.0004            | 0.0030            | 0.0034              | 0.1966              |
|          | Ratios | 25.99             | 26.36             | 2.67                | 2.81                |

TABLE 2

Mean square error and ratios (with respect to the RWM sampler) for EE and limit-EE sampling. Based on 100 replications of 10,000 iterations of each sampler.

partition  $\mathbb{R}^2$  using  $0.5 < 1.0 < 1.75$  so  $\mathcal{X}_1 = \{x \in \mathbb{R}^2 : 0.5x'\Sigma^{-1}x < 0.5\}$ ,  $\mathcal{X}_2, \mathcal{X}_3$  and  $\mathcal{X}_4$  defined similarly. The Means Square Errors and the ratios with respect to RWM is given in Table 3.1. We notice the same drastic loss of efficiency for the EE sampler compared to the limit EE sampler.

Let  $f : \mathcal{X} \rightarrow \mathbb{R}$  be a function such that  $\pi^{(l)}(f) = 0$ . The Monte Carlo estimate of  $\pi^{(l)}(f)$  using the the  $l$ -chain of EE is  $\sum_{k=1}^n f(X_k^{(l)})/n$ . The asymptotics of the EE is more complicated but can be studied along the line of the theory in Section 4. We conjecture that under appropriate conditions, one can obtain (perhaps using similar methods as in this paper) a CLT for  $\sum_{k=1}^n f(X_k^{(l)})/n$  leading to the approximation:

$$\frac{1}{n} \sum_{k=1}^n f(X_k^{(l)}) = \frac{1}{n} \sum_{k=1}^n \pi_k^{(l)}(f) + \kappa_n^{(l)}, \quad (11)$$

where

$$\kappa_n^{(l)} \approx \mathcal{N}\left(0, \frac{\sigma_l^2(f)}{n}\right).$$

where  $\sigma_l^2(f)$  is the asymptotic variance as dictated by the limit kernel  $K^{(l)}$  and  $\frac{1}{n} \sum_{k=1}^n \pi_k^{(l)}(f)$  is the bias term which, as in the case of IR-MCMC, will be of the order  $1/\sqrt{n}$ .

#### 4. Proof of Theorem 2.1

This section is devoted to the proof of Theorem 2.1. Throughout the proof we will use the letter  $C$  to denote a generic finite constant whose actual value might varies from one appearance to another. For  $n \geq 0$ , we call  $\mathcal{F}_n$  the  $\sigma$ -algebra generated by  $\{(X_k^{(0)}, \dots, X_k^{(K)}), k \leq n\}$ . For  $l = 1, \dots, K, n \geq 0$ , we introduce the empirical measure

$$\mu_n^{(l)}(A) = \frac{1}{s_n^{(l)}} \sum_{k=1}^n \omega_k^{(l)} T^{(l)}(X_k^{(l-1)}, A), \quad A \in \mathcal{B}. \quad (12)$$

where  $\omega_k^{(l)} = \omega_l(X_k^{(l-1)})$  and  $s_n^{(l)} = \sum_{j=1}^n \omega_j^{(l)}$ . By construction the dynamics of the algorithm is given by:

$$\Pr(X_n^{(0)} \in A | \mathcal{F}_{n-1}) = P^{(0)}(X_{n-1}^{(0)}, A), \quad A \in \mathcal{B} \quad (13)$$

and for  $l = 1, \dots, K$ ,

$$\Pr(X_n^{(l)} \in A | \mathcal{F}_{n-1}) = \theta_l P^{(l)}(X_{n-1}^{(l)}, A) + (1 - \theta_l) \mu_{n-1}^{(l)}(A), \quad A \in \mathcal{B}. \quad (14)$$

In other words,  $\{X_n^{(0)}, \mathcal{F}_n\}$  is a Markov chain that is assumed  $V$ -uniformly ergodic ((A2)).  $\{X_n^{(0)}\}$  is used to build  $\{\mu_n^{(1)}\}$ , an empirical approximation of  $\pi_1$  which is used in the dynamics of  $\{X_n^{(1)}\}$ . This latter process is in turn used to build  $\{\mu_n^{(2)}\}$ , an empirical approximation of  $\pi_2$  that is used in the dynamics of  $\{X_n^{(2)}\}$ , and so on. The random process thus generated seems complicated, but actually, we will show that some of the main asymptotic properties of  $\{X_n^{(0)}\}$  are in fact inherited by  $\{X_n^{(1)}\}$  and transmitted to  $\{X_n^{(2)}\}$  and transmitted all the way to  $\{X_n^{(K)}\}$ .

The main ingredient of the proof is martingale approximation ([10]). We say that a random process  $\{Y_n, \mathcal{F}_n\}$  admits a martingale approximation if there exist random processes  $\{M_n\}$  and  $\{R_n\}$  such that: (i)  $Y_n = M_n + R_n$  for all  $n \geq 1$  (ii)  $\{M_n, \mathcal{F}_n\}$  is a square-integrable martingale and  $\{R_n\}$  is negligible in some sense (for example  $\sup_n \mathbb{E}[|R_n|^\alpha] < \infty$  for some  $\alpha > 1$ ). When such

decomposition exists, the asymptotics of  $\{Y_n\}$  can be reduced to that of the martingale  $\{M_n\}$ . Below, we will show that such martingale approximation exists for each of the  $\{X_n^{(l)}\}$ .

#### 4.1. Preliminaries results on $P_\mu$

For a probability measure  $\mu$ , a transition kernel  $P$  and  $\theta \in (0, 1)$ , we introduce the transition kernel  $P_\mu$  defined by:

$$P_\mu(x, A) = \theta P(x, A) + (1 - \theta)\mu(A), \quad x \in \mathcal{X}, \quad A \in \mathcal{B}. \quad (15)$$

To study Algorithm 2.1, we will need some basic properties of  $P_\mu$ . This is done in the following three lemmas.

**Lemma 4.1.**  $P_\mu$  is invariant with respect to  $\pi_\mu$  given by:

$$\pi_\mu(A) = (1 - \theta) \sum_{i=0}^{\infty} \theta^i [\mu P^i](A), \quad A \in \mathcal{B}. \quad (16)$$

*Proof.* It is straightforward to check that for all  $A \in \mathcal{B}$ ,  $\int \pi_\mu(dx) P_\mu(x, A) = \pi_\mu(A)$ . We omit the details.  $\square$

**Lemma 4.2.** Assume that  $C_P := \sup_n \|P^n\|_V < \infty$  for some measurable function  $V : (\mathcal{X}, \mathcal{B}) \rightarrow [1, \infty)$ . For probability measures  $\mu_1, \dots, \mu_n$  and  $f \in L_V$ , we have:

$$|P_{\mu_n} \cdots P_{\mu_1} f|_V \leq \left[ \theta^n + (1 - \theta) \sum_{j=0}^{n-1} \theta^j \mu_{j+1}(V) \right] C_P |f|_V. \quad (17)$$

*Proof.* The lemma is a consequence of the following identity which can be easily obtained by recursion:

$$P_{\mu_n} \cdots P_{\mu_1} f = \theta^n P^n f + (1 - \theta) \sum_{j=0}^{n-1} \theta^j \mu_{j+1}(P^j f). \quad (18)$$

$\square$

We introduce the *fundamental kernel*  $U_\mu$  associated to  $P_\mu$  which operates on measurable function  $f : (\mathcal{X}, \mathcal{B}) \rightarrow \mathbb{R}$  as:

$$U_\mu f(x) = \sum_{k=0}^{\infty} [P_\mu^k - \pi_\mu] f(x), \quad x \in \mathcal{X}. \quad (19)$$

Whenever  $U_\mu f$  is well-defined, it gives a solution to the *Poisson equation* for  $P_\mu$ :

$$U_\mu f - P_\mu U_\mu f = f - \pi_\mu(f). \quad (20)$$

In the next lemma, we show that  $U_\mu$  is well defined as long as  $P^n$  does not explode with  $n$ .

**Lemma 4.3.** *Assume that  $C_P := \sup_n \|P^n\|_V < \infty$  for some measurable function  $V : (\mathcal{X}, \mathcal{B}) \rightarrow [1, \infty)$ . For  $f \in L_V$ , we have:*

$$|U_\mu f|_V + |P_\mu U_\mu f|_V \leq C_P |f|_V \frac{1+\theta}{1-\theta} (1 + \mu(V)). \quad (21)$$

Moreover, if  $\mu_1$  and  $\mu_2$  are two probability measures,

$$\sup_{x \in \mathcal{X}} |U_{\mu_1} f(x) - U_{\mu_2} f(x)| + |P_{\mu_1} U_{\mu_1} f(x) - P_{\mu_2} U_{\mu_2} f(x)| \leq C_P |f|_V \frac{1+\theta}{1-\theta} \|\mu_1 - \mu_2\|_V. \quad (22)$$

*Proof.* Let  $f \in L_V$  and  $\mu$  a probability measure. It is fairly simple to check that  $P_\mu^k f = \theta^k P^k f + (1-\theta) \sum_{i=0}^{k-1} \theta^i [\mu P^i](f)$ . Using Lemma 4.1, it follows that  $P_\mu^k [f - \pi_\mu(f)] = \theta^k P^k f - (1-\theta) \sum_{i=k}^{\infty} \theta^i [\mu P^i](f)$ . From which we deduce that

$$U_\mu f = \sum_{k=0}^{\infty} \theta^k P^k f - (1-\theta) \sum_{k=0}^{\infty} \sum_{i=k}^{\infty} \theta^i [\mu P^i](f). \quad (23)$$

Thus:

$$\begin{aligned} \frac{|U_\mu f(x)|}{V(x)} &\leq \sum_{k=0}^{\infty} \theta^k \frac{|P^k f(x)|}{V(x)} + (1-\theta) \sum_{k=0}^{\infty} \sum_{i=k}^{\infty} \theta^i \mu(|P^i f|) \\ &\leq C_P |f|_V \sum_{k=0}^{\infty} \theta^k + (1-\theta) \sum_{k=0}^{\infty} \sum_{i=k}^{\infty} \theta^i \mu\left(\frac{|P^i f|}{V}\right) \\ &\leq C_P |f|_V \sum_{k=0}^{\infty} \theta^k + C_P |f|_V \mu(V) (1-\theta) \sum_{k=0}^{\infty} \sum_{i=k}^{\infty} \theta^i \\ &= \frac{C_P}{1-\theta} |f|_V (1 + \mu(V)). \end{aligned}$$

Similarly,

$$P_\mu U_\mu f = \sum_{k=1}^{\infty} \theta^k P^k f - (1 - \theta) \sum_{k=1}^{\infty} \sum_{i=k}^{\infty} \theta^i [\mu P^i](f), \quad (24)$$

and we proceed as above to show that  $\sup_{|f|_V \leq 1} |P_\mu U_\mu f|_V \leq C_P \frac{\theta}{1-\theta} (1 + \mu(V))$ .

For  $\mu_1$  and  $\mu_2$  given, by using (23), we have:

$$\begin{aligned} |U_{\mu_1} f(x) - U_{\mu_2} f(x)| &\leq (1 - \theta) \sum_{k=0}^{\infty} \sum_{i=k}^{\infty} \theta^i |[\mu_1 - \mu_2](P^i f)| \\ &\leq C_P |f|_V \frac{1}{1 - \theta} \|\mu_1 - \mu_2\|_V. \end{aligned}$$

The argument is the same for  $|P_{\mu_1} U_{\mu_1} f(x) - P_{\mu_2} U_{\mu_2} f(x)|$  using (24).  $\square$

#### 4.2. Existence of martingale approximation for $\{X_n^{(l)}\}$ , $l \geq 1$

To simplify the notations, we denote  $P_n^{(l)}$  the transition kernel  $P_{\mu_n^{(l)}}$  and  $\pi_n^{(l)}$  its invariant distribution as defined in Lemma 4.1. We recall that  $\mu_n^{(l)}(f) = \frac{\sum_{k=1}^n \omega_k^{(l)} T^{(l)} f(X_k^{(l-1)})}{s_n^{(l)}}$ , where  $\omega_k^{(l)} = \omega_l(X_k^{(l-1)})$  and  $s_n^{(l)} = \sum_{j=1}^n \omega_j^{(l)}$ . We denote  $U_n^{(l)}$  the operator  $U_{\mu_n^{(l)}}$  and for  $f : (\mathcal{X}, \mathcal{B}) \rightarrow \mathbb{R}$  measurable, we introduce the partial sum associated to the  $l$ -th chain of the sampler:

$$S_n^{(l)}(f) = \sum_{k=1}^n [f(X_k^{(l)}) - \pi_{k-1}^{(l)}(f)]. \quad (25)$$

The following key result hold. The idea is that the  $l$ -th chain inherits the properties of the  $(l-1)$ -th chain.

**Theorem 4.1.** *Fix  $l \geq 1$ ; assume (A) and the following:*

- a. *For all  $\beta \in [0, 1)$ ,  $\sup_n \mathbb{E} [V^\beta(X_n^{(l-1)})] < \infty$ ;*
- b. *For all  $f \in L_{V^\beta}$ ,  $\beta \in [0, 1)$ ,  $S_n^{(l-1)}(f)$  admits the following martingale approximation:*

$$S_n^{(l-1)}(f) = M_n^{(l-1)}(f) + R_n^{(l-1)}(f), \quad n \geq 1, \quad (26)$$

where  $M_n^{(l-1)}(f) = \sum_{k=1}^n D_n^{(l-1)}(f)$ ,  $\{D_n^{(l-1)}(f), \mathcal{F}_n\}$  is a martingale difference satisfying

$\sup_n \mathbb{E} \left[ \left| D_n^{(l-1)}(f) \right|^\alpha \right] < \infty$ ; and  $\sup_n \mathbb{E} \left[ \left| R_n^{(l-1)}(f) \right|^\alpha \right] < \infty$  for any  $\alpha \geq 1$  such that  $2\alpha\beta < 1$ ;

c. For all  $p \in [1, \infty)$ , and  $\beta \in [0, 1)$ , such that  $\beta p < 1$ ,  $\max_{1 \leq i \leq n} \mathbb{E} \left\{ \frac{\omega_i^{(l)} V^\beta(X_i^{(l-1)})}{s_n^{(l)}} \right\}^p = O\left(\frac{1}{n^p}\right)$ ;

d. For  $f \in L_{V^\beta}$ ,  $\beta \in [0, 1/4)$ , a strong law of large numbers hold:

$$\frac{1}{n} \sum_{k=1}^n f(X_k^{(l-1)}) \xrightarrow{a.s.} \pi^{(l-1)}(f), \quad \text{as } n \rightarrow \infty. \quad (27)$$

Then we have:

i. For all  $\beta \in [0, 1)$ ,  $\sup_n \mathbb{E} \left[ V^\beta(X_n^{(l)}) \right] < \infty$ .

ii. For all  $f \in L_{V^\beta}$ ,  $\beta \in [0, 1)$ ,  $S_n^{(l)}(f)$  admits the following martingale approximation:

$$S_n^{(l)}(f) = M_n^{(l)}(f) + R_n^{(l)}(f), \quad n \geq 1, \quad (28)$$

where  $M_n^{(l)}(f) = \sum_{k=1}^n D_n^{(l)}(f)$ ,  $D_n^{(l)}(f) = U_{n-1}^{(l)} f(X_n^{(l)}) - P_{n-1}^{(l)} U_{n-1}^{(l)} f(X_{n-1}^{(l)})$ ,  $\{D_n^{(l)}(f), \mathcal{F}_n\}$

is a martingale difference,  $\sup_n \mathbb{E} \left[ \left| D_n^{(l)}(f) \right|^\alpha \right] < \infty$  and  $\sup_n \mathbb{E} \left[ \left| R_n^{(l)}(f) \right|^\alpha \right] < \infty$  for any

$\alpha \geq 1$  such that  $2\alpha\beta < 1$ .

iii. For all  $p \in [1, \infty)$ , and  $\beta \in [0, 1)$  such that  $\beta p < 1$ ,  $\max_{1 \leq i \leq n} \mathbb{E} \left\{ \frac{\omega_i^{(l+1)} V^\beta(X_i^{(l)})}{s_n^{(l+1)}} \right\}^p = O\left(\frac{1}{n^p}\right)$ .

iv. For  $f \in L_{V^\beta}$ ,  $\beta \in [0, 1/4)$ , a strong law of large numbers hold:

$$\frac{1}{n} \sum_{k=1}^n f(X_k^{(l)}) \xrightarrow{a.s.} \pi^{(l)}(f), \quad \text{as } n \rightarrow \infty. \quad (29)$$

*Proof.* For the clarity of the presentation, we break the proof in several lemmas. Theorem 4.1 (i) is proved in Lemma 4.4, (ii) is proved in Lemma 4.5, (iii) is proved in Lemma 4.6, and (iv) is proved in Lemma 4.7.  $\square$

**Lemma 4.4.** Under the assumptions of Theorem 4.1, (i) hold.

*Proof.* We start with two remarks. First, it follows from (c.) that  $\mathbb{E} \left\{ \frac{1}{s_n^{(l)}} \sum_{k=1}^n \omega_k^{(l)} V^\beta(X_k^{(l-1)}) \right\}$  is bounded in  $n$  provided  $\beta \in [0, 1)$ . Second, for  $\beta \in [0, 1)$ , we can find a finite constant  $C_\beta$  such

that for all  $f \in L_{V^\beta}$ ,  $i \geq 0$ ,  $n \geq 1$ ,

$$\left| \mathbb{E} \left\{ \mu_n^{(l)} \left( [P^{(l)}]^i f \right) \right\} \right| \leq C_\beta |f|_{V^\beta}. \quad (30)$$

Indeed,

$$\begin{aligned} \left| \mathbb{E} \left\{ \mu_n^{(l)} \left( [P^{(l)}]^i f \right) \right\} \right| &= \left| \mathbb{E} \left\{ \frac{1}{s_n^{(l)}} \sum_{k=1}^n \omega_k^{(l)} T^{(l)} [P^{(l)}]^i f(X_k^{(l-1)}) \right\} \right| \\ &\leq C_1 |f|_{V^\beta} \mathbb{E} \left\{ \frac{1}{s_n^{(l)}} \sum_{k=1}^n \omega_k^{(l)} V^\beta(X_k^{(l-1)}) \right\}, \quad (\text{by (A3)}) \\ &\leq C_\beta |f|_{V^\beta}, \quad (\text{by (c.)}). \end{aligned}$$

Using this (with  $i = 0$  and  $f = V^\beta$ ) and by conditioning on  $\mathcal{F}_{n-1}$ , we see that

$$\begin{aligned} \mathbb{E} [V^\beta(X_n^{(l)})] &= \mathbb{E} [P_{n-1}^{(l)} V^\beta(X_{n-1}^{(l)})] = \mathbb{E} [\theta_l P^{(l)} V^\beta(X_{n-1}^{(l)}) + (1 - \theta_l) \mu_{n-1}^{(l)}(V^\beta)] \\ &\leq \theta_l \mathbb{E} [P^{(l)} V^\beta(X_{n-1}^{(l)})] + (1 - \theta_l) C_\beta \\ &\leq \theta_l^n \mathbb{E} [P^{(l)} V^\beta(X_0^{(l)})] + (1 - \theta_l) C_\beta \sum_{i=0}^{n-1} \theta_l^i, \end{aligned}$$

and given (A3), the right-hand side expression is clearly bounded in  $n$ .  $\square$

**Lemma 4.5.** *Under the assumptions of Theorem 4.1, (ii) hold.*

*Proof.* Take  $f \in L_{V^\beta}$ . Applying (20) to  $P_{k-1}^{(l)}$ , we have  $f - \pi_{k-1}^{(l)}(f) = U_{k-1}^{(l)} f - P_{k-1}^{(l)} U_{k-1}^{(l)} f$ . Using this, we rewrite  $S_n^{(l)}(f)$  as follows.

$$\begin{aligned} S_n^{(l)}(f) &= \sum_{k=1}^n f(X_k^{(l)}) - \pi_{k-1}^{(l)}(f) \\ &= \sum_{k=1}^n U_{k-1}^{(l)} f(X_k^{(l)}) - P_{k-1}^{(l)} U_{k-1}^{(l)} f(X_k^{(l)}) \\ &= \sum_{k=1}^n U_{k-1}^{(l)} f(X_k^{(l)}) - P_{k-1}^{(l)} U_{k-1}^{(l)} f(X_{k-1}^{(l)}) + \sum_{k=1}^n P_{k-1}^{(l)} U_{k-1}^{(l)} f(X_{k-1}^{(l)}) - P_k^{(l)} U_k^{(l)} f(X_k^{(l)}) \\ &\quad + \sum_{k=1}^n P_k^{(l)} U_k^{(l)} f(X_k^{(l)}) - P_{k-1}^{(l)} U_{k-1}^{(l)} f(X_k^{(l)}) \\ &= M_n^{(l)}(f) + R_{n,1}^{(l)}(f) + R_{n,2}^{(l)}(f), \end{aligned} \quad (31)$$

where

$$\begin{aligned}
M_n^{(l)}(f) &= \sum_{k=1}^n U_{k-1}^{(l)} f(X_k^{(l)}) - P_{k-1}^{(l)} U_{k-1}^{(l)} f(X_{k-1}^{(l)}); \\
R_{n,1}^{(l)}(f) &= \sum_{k=1}^n P_{k-1}^{(l)} U_{k-1}^{(l)} f(X_{k-1}^{(l)}) - P_k^{(l)} U_k^{(l)} f(X_k^{(l)}) \\
&= P_0^{(l)} U_0^{(l)} f(X_0^{(l)}) - P_n^{(l)} U_n^{(l)} f(X_n^{(l)}).
\end{aligned} \tag{32}$$

and

$$R_{n,2}^{(l)}(f) = \sum_{k=1}^n P_k^{(l)} U_k^{(l)} f(X_k^{(l)}) - P_{k-1}^{(l)} U_{k-1}^{(l)} f(X_k^{(l)}). \tag{33}$$

Denote  $D_k^{(l)}(f) = U_{k-1}^{(l)} f(X_k^{(l)}) - P_{k-1}^{(l)} U_{k-1}^{(l)} f(X_{k-1}^{(l)})$ .

For  $\alpha \geq 1$  such that  $2\alpha\beta < 1$ , by Minkowski's inequality and Lemma 4.3, we have:

$$\begin{aligned}
\mathbb{E} \left[ \left| D_n^{(l)}(f) \right|^\alpha \right] &\leq C |f|_{V^\beta}^\alpha \left\{ \mathbb{E}^{1/\alpha} \left[ V^{\alpha\beta}(X_n^{(l)}) \left( 1 + \mu_{n-1}^{(l)}(V^\beta) \right)^\alpha \right] \right. \\
&\quad \left. + \mathbb{E}^{1/\alpha} \left[ V^{\alpha\beta}(X_{n-1}^{(l)}) \left( 1 + \mu_{n-1}^{(l)}(V^\beta) \right)^\alpha \right] \right\}^\alpha.
\end{aligned}$$

Now by the Cauchy-Schwartz inequality,

$$\begin{aligned}
\mathbb{E} \left[ V^{\alpha\beta}(X_n^{(l)}) \left( 1 + \mu_{n-1}^{(l)}(V^\beta) \right)^\alpha \right] &\leq \mathbb{E}^{1/2} \left[ V^{2\alpha\beta}(X_n^{(l)}) \right] \mathbb{E}^{1/2} \left[ \left( 1 + \mu_{n-1}^{(l)}(V^\beta) \right)^{2\alpha} \right] \\
&\leq 2^\alpha \mathbb{E}^{1/2} \left[ V^{2\alpha\beta}(X_n^{(l)}) \right] \mathbb{E}^{1/2} \left[ \mu_{n-1}^{(l)}(V^{2\alpha\beta}) \right].
\end{aligned}$$

That  $\sup_n \mathbb{E} \left[ \mu_{n-1}^{(l)}(V^{2\alpha\beta}) \right] < \infty$  was shown in (30) and  $\sup_n \mathbb{E} \left[ V^{2\alpha\beta}(X_n^{(l)}) \right] < \infty$  follows from Lemma 4.4. The same bound hold for  $\mathbb{E} \left[ V^{\alpha\beta}(X_{n-1}^{(l)}) \left( 1 + \mu_{n-1}^{(l)}(V^\beta) \right)^\alpha \right]$ . This implies

$$\sup_n \mathbb{E} \left[ \left| D_n^{(l)}(f) \right|^\alpha \right] < \infty.$$

It is also clear that  $\{D_n^{(l)}(f), \mathcal{F}_n\}$  is a martingale difference.

By the same argument, we see that

$$\sup_n \mathbb{E} \left[ \left| R_{n,1}^{(l)}(f) \right|^\alpha \right] < \infty.$$

To bound  $\mathbb{E} \left[ \left| R_{n,2}^{(l)}(f) \right|^\alpha \right]$ , we note that by Lemma 4.3, we have:

$$\mathbb{E} \left[ \left| P_n^{(l)} U_n^{(l)} f(X_n^{(l)}) - P_{n-1}^{(l)} U_{n-1}^{(l)} f(X_n^{(l)}) \right|^\alpha \right] \leq C |f|_{V^\beta}^\alpha \mathbb{E} \left[ \left\| \mu_n^{(l)} - \mu_{n-1}^{(l)} \right\|_{V^\beta}^\alpha \right]. \quad (34)$$

$\mu_n^{(l)}(f)$  has a recursive expression:  $\mu_n^{(l)}(f) = \mu_{n-1}^{(l)}(f) + \frac{\omega_n^{(l)}}{s_n^{(l)}} \left[ T^{(l)} f(X_n^{(l-1)}) - \mu_{n-1}^{(l)}(f) \right]$ . Using this, we see that

$$\begin{aligned} \left\| \mu_n^{(l)} - \mu_{n-1}^{(l)} \right\|_{V^\beta} &= \sup_{|f|_{V^\beta} \leq 1} \left| \left( \mu_n^{(l)} - \mu_{n-1}^{(l)} \right) (f) \right| \\ &\leq C \frac{\omega_n^{(l)}}{s_n^{(l)}} \left( V^\beta(X_n^{(l-1)}) + \mu_{n-1}^{(l)}(V^\beta) \right) \\ &\leq 2C \frac{\omega_n^{(l)}}{s_n^{(l)}} V^\beta(X_n^{(l-1)}) \mu_{n-1}^{(l)}(V^\beta). \end{aligned}$$

We can then use the Cauchy-Schwartz inequality to get:

$$\begin{aligned} \mathbb{E} \left[ \left\| \mu_n^{(l)} - \mu_{n-1}^{(l)} \right\|_{V^\beta}^\alpha \right] &\leq C \mathbb{E}^{1/2} \left[ \left\{ \frac{\omega_n^{(l)}}{s_n^{(l)}} V^\beta(X_n^{(l-1)}) \right\}^{2\alpha} \right] \mathbb{E}^{1/2} \left[ \left( \mu_{n-1}^{(l)}(V^\beta) \right)^{2\alpha} \right] \\ &\leq C \frac{1}{n^\alpha} \mathbb{E}^{1/2} \left[ \mu_{n-1}^{(l)} \left( V^{2\alpha\beta} \right) \right] \quad (\text{by (c.)}) \\ &= O \left( \frac{1}{n^\alpha} \right), \end{aligned}$$

since  $\sup_n \mathbb{E} \left[ \mu_{n-1}^{(l)} \left( V^{2\alpha\beta} \right) \right] < \infty$  as shown in (30). This bound together with (34) implies:

$$\mathbb{E} \left[ \left| P_n^{(l)} U_n^{(l)} f(X_n^{(l)}) - P_{n-1}^{(l)} U_{n-1}^{(l)} f(X_n^{(l)}) \right|^\alpha \right] \leq C |f|_{V^\beta}^\alpha \frac{1}{n^\alpha}. \quad (35)$$

Applying Minkowski's inequality to (33), it follows from (35) that:

$$\sup_n \mathbb{E} \left[ \left| R_{n,2}^{(l)}(f) \right|^\alpha \right] < \infty,$$

which completes the proof. □

**Lemma 4.6.** *Under the assumptions of Theorem 4.1, (iii) hold.*

*Proof.* Write  $V_{i,\beta}^{(l)} = V^\beta(X_i^{(l)})$ . Let  $c$  as in (A1) and  $\kappa \in (0, c)$ . For  $p \geq 1$  and  $1 \leq i \leq n$  we have:

$$\begin{aligned} \mathbb{E} \left[ \frac{\omega_i^{(l+1)} V_{i,\beta}^{(l)}}{s_n^{(l+1)}} \right]^p &= \mathbb{E} \left[ \frac{\omega_i^{(l+1)} V_{i,\beta}^{(l)}}{s_n^{(l+1)}} \mathbf{1}_{\{s_n^{(l+1)} \geq \sum_{j=1}^n (\pi_{j-1}^{(l)}(\omega_{l+1}) - \kappa)\}} \right]^p \\ &\quad + \mathbb{E} \left[ \frac{\omega_i^{(l+1)} V_{i,\beta}^{(l)}}{s_n^{(l+1)}} \mathbf{1}_{\{s_n^{(l+1)} < \sum_{j=1}^n (\pi_{j-1}^{(l)}(\omega_{l+1}) - \kappa)\}} \right]^p. \end{aligned}$$

Since  $\omega_{l+1}(x) \geq c$  for all  $x \in \mathcal{X}$  and  $\kappa \in (0, c)$ ,  $\pi_{j-1}^{(l)}(\omega_{l+1}) - \kappa \geq c - \kappa$ , and

$$\begin{aligned} \mathbb{E} \left[ \frac{\omega_i^{(l+1)} V_{i,\beta}^{(l)}}{s_n^{(l+1)}} \mathbf{1}_{\{s_n^{(l+1)} \geq \sum_{j=1}^n (\pi_{j-1}^{(l)}(\omega_{l+1}) - \kappa)\}} \right]^p &\leq \frac{1}{n^p (c - \kappa)^p} \mathbb{E} \left[ \omega_i^{(l+1)} V_{i,\beta}^{(l)} \right]^p \\ &\leq C \frac{\mathbb{E} \left( V^{\beta p}(X_i^{(l)}) \right)}{n^p (c_l - \kappa)^p} \\ &= O\left(\frac{1}{n^p}\right), \end{aligned}$$

by Lemma 4.4.

Let  $q_1, q_2 \in (1, \infty)$  be such that  $\beta p q_1 < 1$  and  $1/q_1 + 1/q_2 = 1$ . Using Holder's inequality, we can bound the second term as follows:

$$\begin{aligned} \mathbb{E} \left[ \frac{\omega_i^{(l+1)} V_{i,\beta}^{(l)}}{s_n^{(l+1)}} \mathbf{1}_{\{s_n^{(l+1)} < \sum_{j=1}^n (\pi_{j-1}^{(l)}(\omega_{l+1}) - \kappa)\}} \right]^p &\leq \mathbb{E} \left[ V^{\beta p}(X_i^{(l)}) \mathbf{1}_{\{s_n^{(l+1)} < \sum_{j=1}^n (\pi_{j-1}^{(l)}(\omega_{l+1}) - \kappa)\}} \right] \\ &\leq \mathbb{E}^{1/q_1} \left[ V^{\beta p q_1}(X_i^{(l)}) \right] \left( \Pr \left[ \frac{1}{n} S_n^{(l)}(\omega_{l+1}) < -\kappa \right] \right)^{1/q_2} \\ &\leq \frac{C}{n^{2p} \kappa^{2p}} \mathbb{E}^{1/q_2} \left[ \left| S_n^{(l)}(\omega_{l+1}) \right|^{2q_2 p} \right]. \end{aligned}$$

For the last inequality, we use Lemma 4.4, to bound  $\mathbb{E}^{1/q_1} \left[ V^{\beta p q_1}(X_i^{(l)}) \right]$  and Markov's inequality.

The proof will be finished if we show that  $\mathbb{E} \left[ \left| S_n^{(l)}(\omega_{l+1}) \right|^{2q_2 p} \right] = O(n^{q_2 p})$ .

By Lemma 4.5, and since  $\omega_{l+1}$  is a bounded function,  $\{S_n^{(l)}(\omega_{l+1})\}$  admits a martingale approximation  $S_n^{(l)}(\omega_{l+1}) = M_n^{(l)} + R_n^{(l)}$ , where  $\sup_n \mathbb{E} \left[ \left| R_n^{(l)} \right|^\alpha \right] < \infty$  for all  $\alpha \geq 1$ ;  $M_n^{(l)} = \sum_{k=1}^n D_n^{(l)}$  and  $\sup_n \mathbb{E} \left[ \left| D_n^{(l)} \right|^\alpha \right] < \infty$  for all  $\alpha \geq 1$ . So to show that  $\mathbb{E} \left[ \left| S_n^{(l)}(\omega_{l+1}) \right|^{2q_2 p} \right] = O(n^{q_2 p})$ , it is sufficient to show that  $\mathbb{E} \left[ \left| M_n^{(l)} \right|^{2q_2 p} \right] = O(n^{q_2 p})$ . This follows essentially from Burkholder's inequality for

martingale. Indeed, we have:

$$\begin{aligned}
\mathbb{E} \left[ \left| M_n^{(l)} \right|^{2q_{2p}} \right] &\leq \mathbb{E} \left[ \left\{ \sum_{k=1}^n \left| D_k^{(l)} \right|^2 \right\}^{q_{2p}} \right], \quad (\text{ by Burkholder's inequality }) \\
&\leq \left\{ \sum_{k=1}^n \mathbb{E}^{1/q_{2p}} \left[ \left| D_k^{(l)} \right|^{2q_{2p}} \right] \right\}^{q_{2p}}, \quad (\text{ by Minkowski's inequality }) \\
&= O(n^{q_{2p}}), \quad (\text{ by Lemma 4.5}).
\end{aligned}$$

□

**Lemma 4.7.** *Under the assumptions of Theorem 4.1, (iv) hold.*

*Proof.* For  $f \in L_{V^\beta}$ ,  $\beta < 1/4$ , it follows from the martingale approximation obtained in Lemma 4.5 that:  $S_n^{(l)}(f) = M_n^{(l)}(f) + R_n^{(l)}(f)$ , where  $M_n^{(l)}(f) = \sum_{k=1}^n D_n^{(l)}(f)$ ,  $\sup_n \mathbb{E} \left[ \left| D_n^{(l)}(f) \right|^2 \right] < \infty$  and  $\sup_n \mathbb{E} \left[ \left| R_n^{(l)}(f) \right|^2 \right] < \infty$ . Clearly,  $\sup_n \mathbb{E} \left[ \left| R_n^{(l)}(f) \right|^2 \right] < \infty$  implies that  $R_n^{(l)}(f)/n \rightarrow 0$  almost surely. And by the strong law of large numbers for martingale ([8]),  $\sup_n \mathbb{E} \left[ \left| D_n^{(l)}(f) \right|^2 \right] < \infty$  implies that  $M_n^{(l)}(f)/n \rightarrow 0$  almost surely. Thus,  $S_n^{(l)}(f)/n \rightarrow 0$  almost surely. It only remains to show that  $\pi_n^{(l)}(f) \rightarrow \pi^{(l)}(f)$  as  $n \rightarrow \infty$  and the lemma will be proved.

$\pi_n^{(l)}(f) = (1 - \theta_l) \sum_{i=0}^{\infty} \theta_l^i \left[ \mu_n^{(l)} \left( [P^{(l)}]^i f \right) \right]$ . By the strong law of large number for  $\{X_n^{(l-1)}\}$  assumed in Theorem 4.1 (d.),

$$\mu_n^{(l)} \left( [P^{(l)}]^i f \right) = \frac{\frac{1}{n} \sum_{k=1}^n \omega_l(X_k^{(l-1)}) T^{(l)} [P^{(l)}]^i f(X_k^{(l-1)})}{\frac{1}{n} \sum_{k=1}^n \omega_l(X_k^{(l-1)})} \rightarrow \frac{\pi^{(l-1)} \left( \omega_l T^{(l)} [P^{(l)}]^i f \right)}{\pi^{(l-1)}(\omega_l)} = \pi_l(f),$$

as  $n \rightarrow \infty$ . Moreover,  $\left| \mu_n^{(l)} \left( [P^{(l)}]^i f \right) \right| \leq C \|f\|_{V^\beta} \mu_n^{(l)} \left( V^\beta \right) \rightarrow C \|f\|_{V^\beta} \pi^{(l-1)}(V^\beta) < \infty$ . Therefore for almost every sample path, we can find a finite constant  $C'$  (that does not depend on  $n$  or  $i$ ) such that  $\left| \mu_n^{(l)} \left( [P^{(l)}]^i f \right) \right| \leq C'$ . By Lebesgue dominated theorem,  $\pi_n^{(l)}(f) \rightarrow \pi^{(l)}(f)(1 - \theta_l) \sum \theta_l^i = \pi^{(l)}(f)$ . □

### 4.3. Asymptotic results for $\{X_n^{(0)}\}$

Since  $\{X_n^{(0)}\}$  is assumed  $V$ -uniformly ergodic, it has a nice asymptotic behavior. The next proposition summarizes some of these properties that will be needed to initialize the argument by recursion behind Theorem 4.1. For  $f \in L_V$ , we introduce the partial sum:

$$S_n^{(0)}(f) = \sum_{k=1}^n \left[ f(X_k^{(0)}) - \pi^{(0)}(f) \right], \quad n \geq 1. \quad (36)$$

**Proposition 4.1.** *Assume that (A) hold. Then:*

- i.  $\sup_n \mathbb{E} \left[ V(X_n^{(0)}) \right] < \infty$ .
- ii. For all  $f \in L_{V^\beta}$ ,  $\beta \in [0, 1]$ ,  $S_n^{(0)}(f)$  admits the following martingale approximation:

$$S_n^{(0)}(f) = M_n^{(0)}(f) + R_n^{(0)}(f), \quad n \geq 1, \quad (37)$$

where  $M_n^{(0)}(f) = \sum_{k=1}^n D_k^{(0)}(f)$ ,  $\{D_n^{(0)}(f), \mathcal{F}_n\}$  is a martingale difference satisfying

$\sup_n \mathbb{E} \left[ |D_n^{(0)}(f)|^\alpha \right] < \infty$  and  $\sup_n \mathbb{E} \left[ |R_n^{(0)}(f)|^\alpha \right] < \infty$  for any  $\alpha \geq 1$  such that  $\alpha\beta \leq 1$ .

- iii. For all  $p \in [1, \infty)$ , and  $\beta \in [0, 1]$  such that  $\beta p \leq 1$ ,  $\max_{1 \leq i \leq n} \mathbb{E} \left\{ \frac{\omega_i^{(1)} V^\beta(X_i^{(0)})}{s_n^{(1)}} \right\}^p = O\left(\frac{1}{n^p}\right)$ .
- iv. For  $f \in L_{V^{1/2}}$ ,

$$\frac{1}{n} \sum_{k=1}^n f(X_k^{(0)}) \xrightarrow{a.s.} \pi^{(0)}(f), \quad \text{as } n \rightarrow \infty.$$

*Proof.* The proof uses essentially the same techniques as in the proof of Theorem 4.1 with the important simplification here that  $\{X_n^{(0)}\}$  is a Markov chain. We omit the details.  $\square$

### 4.4. Proof of Theorem 2.1 (i)

Follows immediately from Proposition 4.1 and Theorem 4.1.

#### 4.5. Proof of Theorem 2.1 (ii)

Let  $f \in L_{V^\beta}$ ,  $\beta \in [0, 1/8)$ . As before, the starting point of the proof is the martingale approximation:  $S_n^{(l)}(f) = M_n^{(l)}(f) + R_n^{(l)}(f)$ ,  $M_n^{(l)}(f) = \sum_{k=1}^n D_k^{(l)}(f)$ . Since  $\sup_n \mathbb{E} \left[ \left| R_n^{(l)}(f) \right|^2 \right] < \infty$ ,  $R_n^{(l)}(f)/n \rightarrow 0$  in probability. Now we need to show that the martingale  $\{M_n^{(l)}(f)\}$  satisfies a central limit theorem (CLT). To simplify the notations, we will omit the upper script  $l$  in the rest of the proof. For example, we will write  $D_k(f)$  instead of  $D_k^{(l)}(f)$ ;  $P$  instead of  $P^{(l)}$ ;  $\pi$  instead of  $\pi^{(l)}$  etc...

We apply Brown's CLT for martingales ([7] Theorem 2). We need to prove that the Lindeberg condition hold:

$$\frac{1}{n} \sum_{k=1}^n \mathbb{E} \left[ D_k^2(f) \mathbf{1}_{\{|D_k(f)| > \varepsilon \sqrt{n}\}} \right] \xrightarrow{P} 0, \quad \text{for all } \varepsilon > 0 \text{ as } n \rightarrow \infty. \quad (38)$$

and that

$$\frac{1}{n} \sum_{k=1}^n \mathbb{E}(D_k^2(f) | \mathcal{F}_{k-1}) \rightarrow \sigma^2(f), \quad (39)$$

where  $\sigma^2(f) = \pi(\bar{f}^2) + 2 \sum_{i=1}^{\infty} \theta^i \pi[\bar{f} P^i \bar{f}]$ ,  $\bar{f} = f - \pi(f)$ .

Since  $\beta < 1/8$ , the Lyapunov condition  $\sup_n \mathbb{E}(|D_n(f)|^{2+\varepsilon}) < \infty$  follows from Theorem 4.1 (ii) and it implies the Lindeberg condition (38).

Without any loss of generality, we can assume that  $\pi(f) = 0$ . To show (39), we introduce the transition kernel  $P^* = \theta P + (1 - \theta)\pi$ . Note that  $\pi P^* = \pi$ . Associated with  $P^*$  is the operator  $Uf = \sum_{k=0}^{\infty} [P^*]^k f = \sum_{k=0}^{\infty} \theta^k P^k f$ . Clearly,  $Uf$  satisfies the Poisson equation  $Uf - P^*(Uf) = f$ .

$$\mathbb{E} \left( D_n^2(f) | \mathcal{F}_{n-1} \right) = P_{n-1} [U_{n-1} f]^2 (X_{n-1}) - (P_{n-1} U_{n-1} f (X_{n-1}))^2 \quad (40)$$

$$= \left[ P^*(Uf)^2 - (P^* Uf)^2 \right] (X_{n-1}) + \delta_{n-1}, \quad (41)$$

where  $\delta_{n-1} = P_{n-1} [U_{n-1}f]^2(X_{n-1}) - P^*(Uf)^2(X_{n-1}) - (P_{n-1}U_{n-1}f(X_{n-1}))^2 + (P^*Uf)^2(X_{n-1})$ .

Since  $f \in L_{V^\beta}$ ,  $Uf \in L_{V^\beta}$  by (A3). Thus  $P^*(Uf)^2 - (P^*Uf)^2 \in L_{V^{2\beta}}$  and the strong law of large numbers of Theorem 2.1 (i) applies and gives that  $\frac{1}{n} \sum_{i=1}^n [P^*(Uf)^2 - (P^*Uf)^2](X_{i-1}) \rightarrow \pi [P^*(Uf)^2 - (P^*Uf)^2]$  a.s. as  $n \rightarrow \infty$ . Since  $Uf = f + P^*Uf$ , it follows that  $(Uf)^2 - (P^*Uf)^2 = -f^2 + 2fUf$  which implies that

$$\begin{aligned} \pi [P^*(Uf)^2 - (P^*Uf)^2] &= \pi [(Uf)^2 - (P^*Uf)^2] \\ &= -\pi(f^2) + 2\pi[fUf] \\ &= \pi(f^2) + 2 \sum_{i=1}^{\infty} \theta^i \pi[fP^i f]. \end{aligned}$$

It is now clear that the proof will be finished once we prove that  $\frac{1}{n} \sum_{k=1}^n \delta_{k-1} \rightarrow 0$  as  $n \rightarrow \infty$ , Pr-a.s. Denote  $r_n(f) = \sum_{k=0}^{\infty} \sum_{i=k}^{\infty} \theta^i \mu_n(P^i f)$  so that  $U_n f = Uf - (1 - \theta)r_n(f)$ . Also note that we can rewrite  $P_n$  as  $P_n = P^* + (1 - \theta)[\mu_n - \pi]$ . Then a straightforward calculus gives:

$$\begin{aligned} \delta_{n-1} &= P_{n-1} [U_{n-1}f]^2(X_{n-1}) - P^*(Uf)^2(X_{n-1}) - (P_{n-1}U_{n-1}f(X_{n-1}))^2 + (P^*Uf)^2(X_{n-1}) \\ &= (1 - \theta)(\mu_{n-1} - \pi) [(Uf)^2] - (1 - \theta)^2 [(\mu_{n-1} - \pi)(Uf)]^2 \\ &\quad - 2(1 - \theta)(\mu_{n-1} - \pi)(Uf)P^*Uf(X_{n-1}) + (1 - \theta)^3 r_n(f)(\mu_{n-1} - \pi)(Uf). \end{aligned}$$

By the SLLN,  $(\mu_{n-1} - \pi) [(Uf)^2]$  and  $(\mu_{n-1} - \pi)(Uf)$  converge almost surely to 0. Since  $\mu_n(P^i f)$  also converges to 0, it follows that  $r_n(f)$  is bounded and therefore,  $r_n(f)(\mu_{n-1} - \pi)(Uf)$  converges to 0. For the last term  $(\mu_{n-1} - \pi)(Uf)P^*Uf(X_{n-1})$ , we note that by the SLLN,  $\frac{1}{n} \sum_{k=1}^n P^*|Uf|(X_k)$  converges a.s. to some finite constant and  $(\mu_n - \pi)(Uf)$  converges to 0, therefore,

$\frac{1}{n} \sum_{k=1}^n [(\mu_k - \pi)(Uf)] P^*Uf(X_k)$  converges a.s. 0 and we are finished.

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